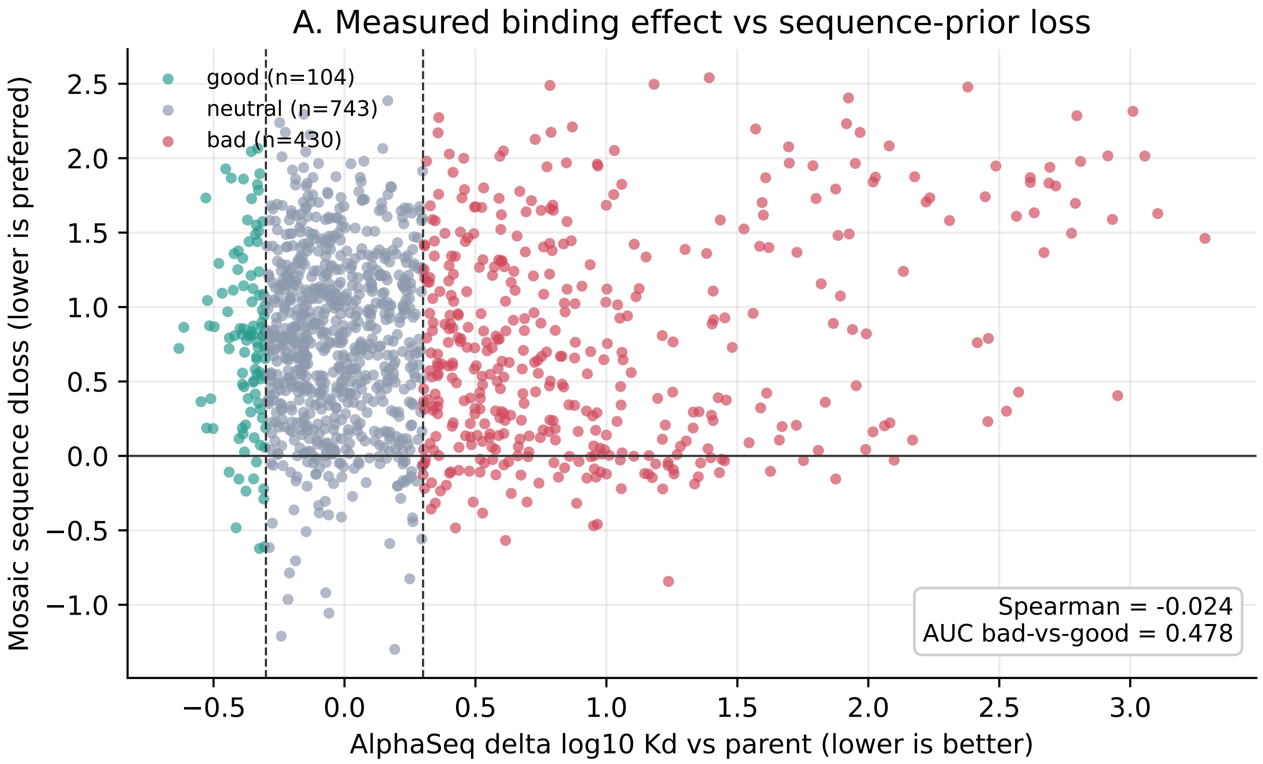
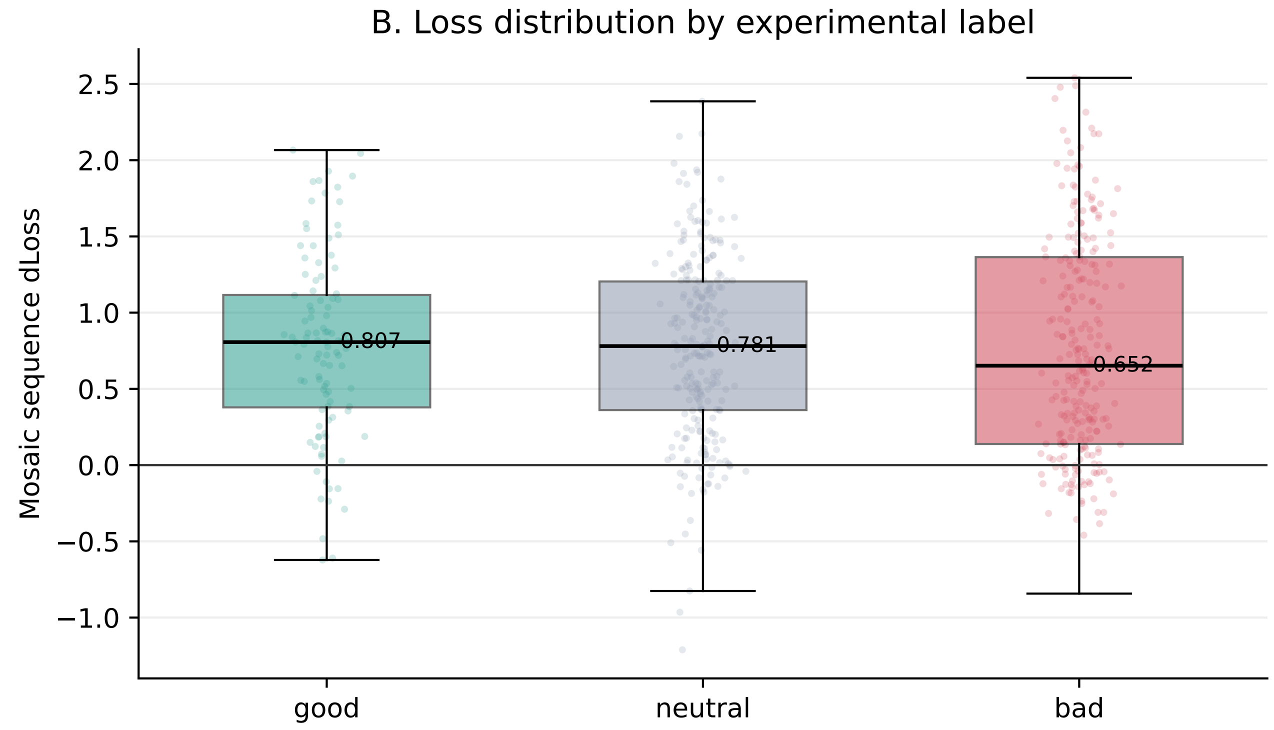
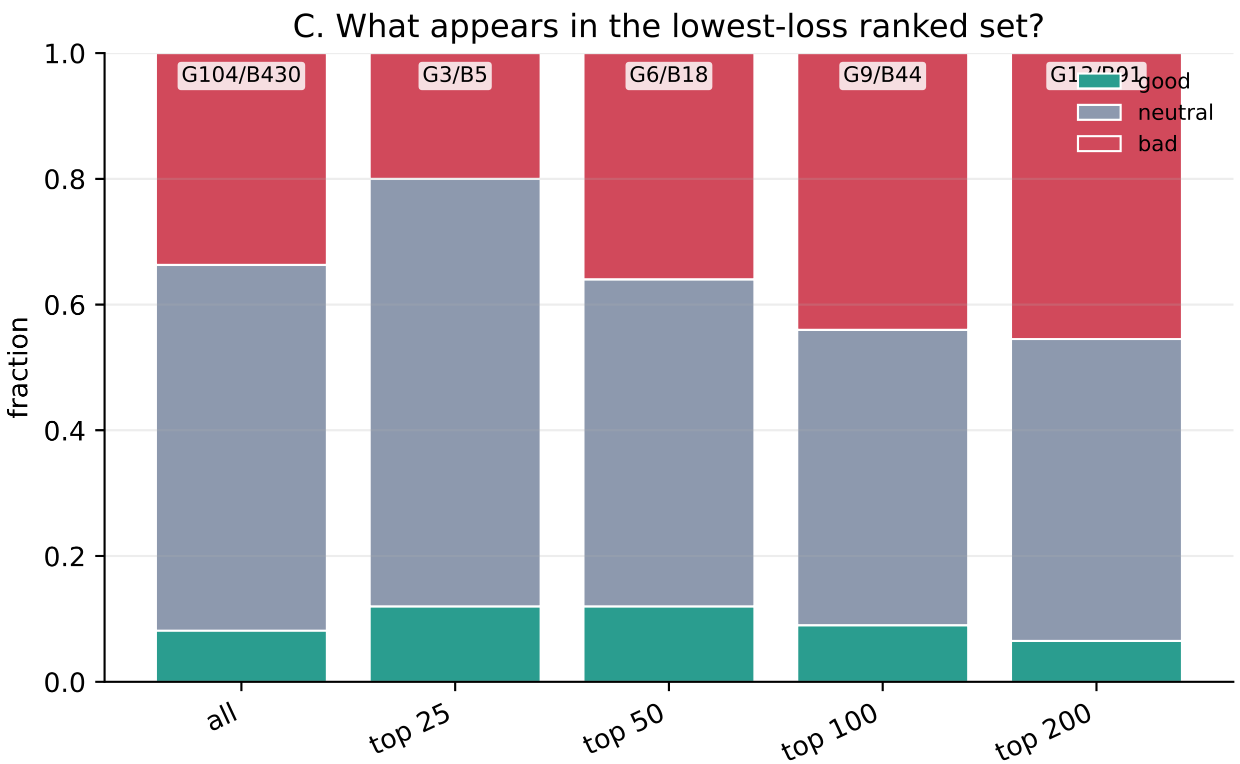
AlphaSeq SARS-CoV-2 RBD VHH DMS: ESM2/AbLang2 priors do not identify binding-good mutations







D. Discordant examples

Sequence priors are useful regularizers, but here they are not a binding oracle. Several disruptive mutations receive low model loss, while several measured beneficial mutations are penalized.

Model likes, AlphaSeq says bad

mutation	AlphaSeq d	model dLoss
F47W	+1.237	-0.843
W53R	+0.615	-0.568
W53V	+0.424	-0.484
D111Y	+0.951	-0.470
W53E	+0.966	-0.459
K87T	+0.527	-0.384

Experiment says good, model penalizes

mutation	AlphaSeq d	model dLoss
A92Y	-0.331	+2.066
D62W	-0.356	+2.045
T91H	-0.454	+1.928
A92E	-0.323	+1.896
D73F	-0.432	+1.866
R67M	-0.386	+1.860

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n mutations: 1277
good/bad: 104/430
top100 good/bad: 9/44
median loss good: 0.807
median loss bad:  0.652
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Parent VHH background from AlphaSeq SARS-CoV-2 RBD; good/bad labels use delta log10 Kd thresholds -0.3/+0.3 relative to parent.